

Linux Commands

File System Commands:

Command	Description	Common Options / Variations	Example	Output / Use Case
pwd	Prints the full path of your current working directory.	(none)	pwd	/home/user → shows where you are currently located in the filesystem.
ls	Lists files and directories in the current folder.	-l (detailed list), -a (show hidden files), -h (human-readable sizes), -R (recursive listing)	ls -lah	Displays all files (including hidden), with permissions, size, and modification date.
cd	Changes your current directory.	cd .. (go up one level), cd ~ (home dir), cd - (previous dir)	cd /var/log	Moves you to /var/log directory. Useful for navigation.
mkdir	Creates new directories (folders).	-p (creates parent directories automatically)	mkdir -p /tmp/practice/test	Creates /tmp/practice/test. The -p prevents errors if parent dirs don't exist.
rmdir	Removes an empty directory.	(none)	rmdir old_dir	Deletes old_dir if it has no files. Use rm -r for non-empty directories.
touch	Creates an empty file or updates its timestamp.	(none)	touch notes.txt	Creates notes.txt file if it doesn't exist. Commonly used to create temp files.
rm	Deletes files or directories.	-r (recursive), -f (force), -i (confirm before delete)	rm -rf test_folder	Deletes a directory and all its contents. Be careful — no undo!
cat	Displays contents of a file or combines multiple files.	(none)	cat file1.txt or cat file1.txt file2.txt > combined.txt	Reads file contents or merges multiple files into one.
more	Displays text files one page at a time (scroll forward only).	(none)	more bigfile.txt	Used for viewing long text outputs page-by-page. Press Space to move down.
less	Advanced file viewer — allows scrolling up/down and searching	/pattern (search), G (go to end), g (top), q (quit)	less /var/log/syslog	Used to view large logs interactively. Very common in DevOps.

	inside file.			
find	Searches for files and directories based on name, size, type, or modification date.	-name, -type, -size, -mtime, -exec	find . -name "*.log" or find /var/log -size +10M	Finds .log files or files larger than 10 MB — essential for log management and cleanup scripts.

File Permission Commands:

Command	Description	Variants	Example	Output / Use Case
chmod	Change file permissions (read/write/execute)	+x, -r, numeric 644, 755	chmod 755 script.sh	Assigns execute permission to owner, read/write to owner, read to group/others.
chown	Change file owner	user:group	chown root:root file.txt	Changes file ownership to root user and group.
chgrp	Change file group	group name	chgrp dev file.txt	Assigns the file to a different group.
umask	Set default permissions for new files	numeric e.g., 022, 027	umask 027	New files get default permissions according to umask.
ls -l	List files with permissions, owner, group	-a (hidden), -h (human-readable)	ls -lh /tmp	Shows detailed file info: permissions, owner, size, date.

Users & Groups:

useradd	Add a new user	-m (create home), -s (set shell)	sudo useradd -m abdullah	Creates a new user with home directory.
usermod	Modify user account	-aG group (add to group)	sudo usermod -aG sudo testuser	Adds testuser to sudo group.
passwd	Set or change user password	(none)	sudo passwd testuser	Sets password for user; login requires this password.
groupadd	Create a new group	(none)	sudo groupadd devops	Creates a group named devops.
whoami	Show current logged-in user	(none)	whoami	Displays current user.

File Operation Commands:

Command	Description	Variants	Example	Output / Use Case
cp	Copy files/directories	-r (recursive), -v (verbose)	cp file1.txt backup/	Copies file or folder; verbose shows progress.
mv	Move or rename files/directories	(none)	mv old.txt new.txt	Renames a file or moves it to another directory.
grep	Search text patterns in files	-i (ignore case), -r (recursive), --color	grep "error" /var/log/syslog	Finds lines containing "error" in log files.
diff	Compare two files line by line	-u (unified)	diff file1.txt file2.txt	Shows differences between files.
sort	Sort file contents	-r (reverse), -n (numeric)	sort names.txt	Outputs file sorted alphabetically or numerically.
uniq	Remove duplicate lines (must be sorted first)	-c (count occurrences)	`sort list.txt	uniq -c`
wc	Count lines, words, characters	-l, -w, -c	wc -l file.txt	Shows number of lines in file.
cut	Extract columns/fields from text	-d (delimiter), -f (field)	cut -d":" -f1 /etc/passwd	Extracts usernames from /etc/passwd.
awk	Pattern scanning & text processing	{print \$1}, -F (field separator)	awk -F: '{print \$1}' /etc/passwd	Prints first column of CSV or /etc/passwd.
sed	Stream editor — text replacement	s/old/new/g	sed 's/test/abc/g' file.txt	Replaces all occurrences of "test" with "abc".

File Structure:

Command	Description	Variants	Example	Output / Use Case
/home	Home directories of users	(none)	ls /home	Shows user home folders.
/etc	System configuration files	(none)	ls /etc	Contains config files like passwd, group.
/var	Variable data: logs, databases	(none)	ls /var/log	View logs with less or tail.
/usr	User applications & binaries	(none)	ls /usr/bin	Installed binaries and programs reside here.
/bin	Essential system binaries	(none)	ls /bin	Basic commands like ls, cp, mv.
/opt	Optional 3rd-party software	(none)	ls /opt	Software installed manually by

				users/admins.
/tmp	Temporary files	(none)	ls /tmp	Safe place to practice creating temporary files.